

COSC 594

Interplay between RL and GenAI

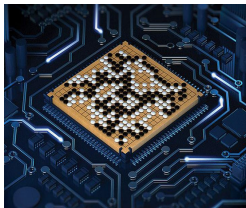
**(Reinforcement Learning
Generative Artificial Intelligence)**

Dongsheng Ding

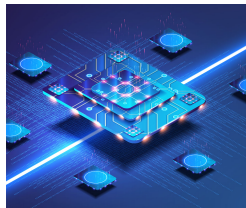
Department of Electrical Engineering and Computer Science
University of Tennessee, Knoxville

Success stories of RL

■ HUMAN-LEVEL PERFORMANCE



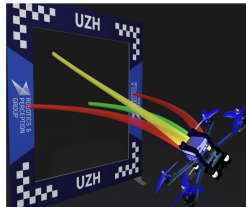
AlphaGo '16



Chip Placement '21

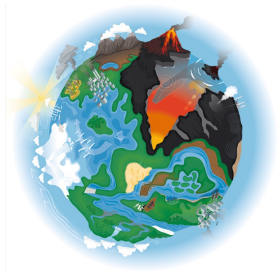


GT Sophie '22



Drone Racing '23

RL framework



environment

information



decision



agent

RL framework



environment

information



decision



agent



reward

goal

RL framework



environment

information



decision



agent



reward

goal

Reinforcement learns
human-level skills by trial and error

Success stories of GenAI

■ HUMAN-LEVEL CONTENT



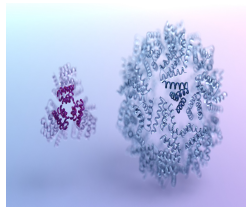
Copilot '23



DALL-E 3 '23

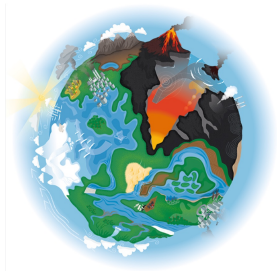


Sora '24



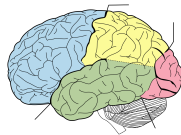
AlphaFold3 '24

GenAI framework



environment

information



agent

GenAI framework



environment

information



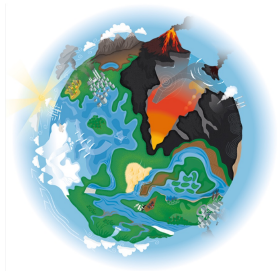
agent



dream

likelihood

GenAI framework



environment

information



agent



dream

likelihood

Generation learns
human-level skills by imagination

RL vs. GenAI

■ RL

- ★ dynamic environment

■ GenAI

- ★ static environment

RL vs. GenAI

■ RL

- ★ dynamic environment
- ★ interactive

■ GenAI

- ★ static environment
- ★ non-interactive

RL vs. GenAI

■ RL

- ★ dynamic environment
- ★ interactive
- ★ reward maximization

■ GenAI

- ★ static environment
- ★ non-interactive
- ★ likelihood maximization

RL vs. GenAI

■ RL

- ★ dynamic environment
- ★ interactive
- ★ reward maximization
- ★ CNN

■ GenAI

- ★ static environment
- ★ non-interactive
- ★ likelihood maximization
- ★ Transformer or U-Net

RL vs. GenAI

■ RL

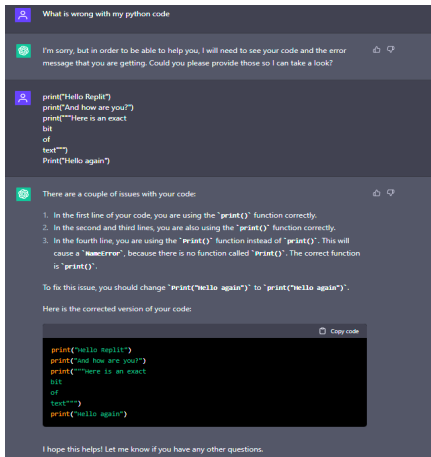
- ★ dynamic environment
- ★ interactive
- ★ reward maximization
- ★ CNN
- ★ autonomous systems

■ GenAI

- ★ static environment
- ★ non-interactive
- ★ likelihood maximization
- ★ Transformer or U-Net
- ★ content generation

RL for GenAI

■ NATURAL LANGUAGE GENERATION



RL improves ChatGPT's responses.

RL for GenAI

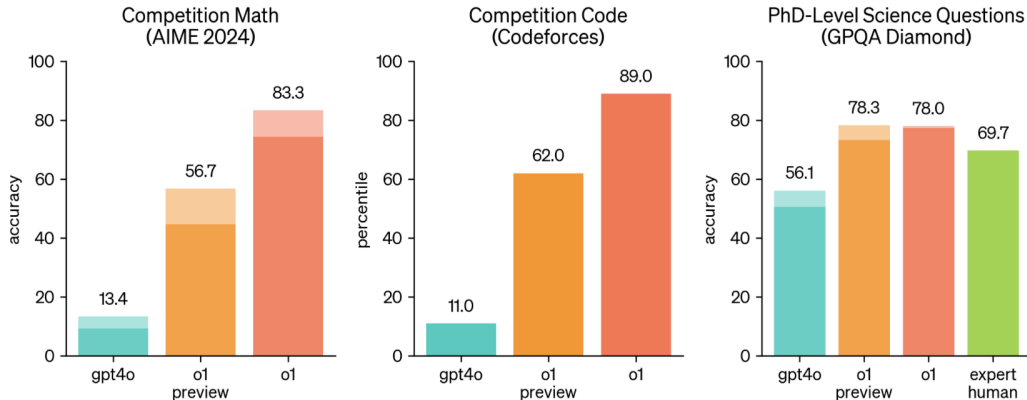
■ IMAGE GENERATION



RL helps Dall E to generate realistic images.

RL for GenAI

REASONING



RL helps ChatGPT solve math/code/science problems.

RL improves GenAI by optimizing for user requirements.

GenAI for RL

■ SYNTHETIC DATA GENERATION



Generative scenes from NVIDIA Isaac Sim speed up robotic training.

GenAI for RL

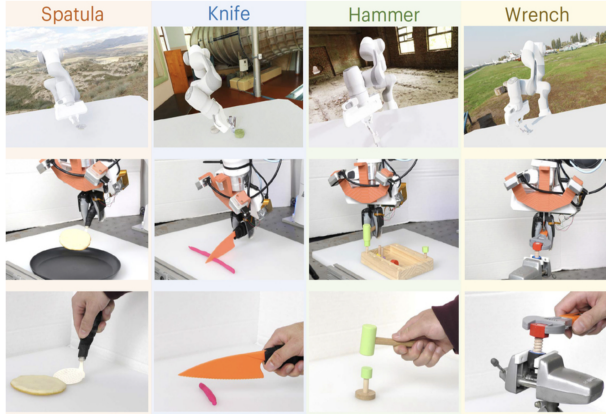
■ PATIENT SIMULATION



Generative patient simulations (Memrizz AI Patient) help train treatment policies.

GenAI for RL

■ MULTIPURPOSE ROBOT



Combined generative models help robots learn better (MIT News).

GenAI reduces RL sample complexity and safety risks.

This course focuses on their interaction
(**Optimization/Learning problems, algorithms, and analyses**)

Part I

RL for GenAI

(**Generative language models and diffusion models**)

Part II

GenAI for RL

(**Policy optimization**)

Outline

- INTRODUCTION OF RL AND GenAI
- ALIGNMENT AND COMPOSITION OF LANGUAGE MODELS
- ALIGNMENT AND COMPOSITION OF DIFFUSION MODELS
- UNLEARNING IN LANGUAGE AND DIFFUSION MODELS
- DIFFUSION MODELS FOR POLICY OPTIMIZATION
- DIFFUSION MODELS FOR ROBOTICS
- OTHER TOPICS

Logistics

■ 10% CLASS PARTICIPATION

Class participation includes asking questions, engaging in discussions, scribing lectures, and providing feedback.

■ 40% HOMEWORK (4 ASSIGNMENTS, EACH 10%)

There are four homework assignments that are equally weighted. Discussions with others are allowed, but all answers must be completed independently.

■ 50% PROJECT (20% PRESENTATION AND 30% REPORT)

Project aims to present and discuss a machine learning paper on the interplay between RL and GenAI. Students will work in pairs to present the paper and summarize it in a joint report in $\text{\LaTeX}$. The paper and presentation date will be assigned.

Prerequisites

■ LINEAR ALGEBRA AND CALCULUS

Matrix inverse, rank, eigenvalue; gradient, Lipschitz continuity, smoothness

■ PROBABILITY

Random variable, probability distribution, expectation, conditional expectation

■ CONVEX ANALYSIS

Convex function, optimality condition, gradient descent, Lagrangian, duality

■ MACHINE LEARNING

Supervised learning, least squares

Course materials

■ SLIDES

Course slides will be shared in Canvas.

■ READINGS

References will be shared in Canvas.

■ HOMEWORKS AND PROJECTS

Homework assignments will be announced on Canvas. Project assignments will also be communicated through Canvas.

Feedback

**Any suggestions on lectures, homeworks, and projects
are welcome.**

(Email: dongshed@utk.edu)